Customer Shopping Behaviour Analysis

1. Project Overview

This project analyses customer shopping behaviour using transactional data from approximately **3,900 purchases** across multiple product categories. The objective is to identify key insights related to **spending patterns, customer segmentation, product preferences, and subscription behaviour** to support data-driven business decisions.

2. Tools Used

• Python (Pandas, NumPy) – Data cleaning, preprocessing, and feature engineering

• SQL (MySQL) – Data analysis and business query execution

• Power BI – Interactive dashboard creation and data visualization

3. Dataset Summary

- **Total Records:** 3,900
- **Total Columns:** 18

Key Features Include:

- **Customer Demographics:** Age, Gender, Location, Subscription Status
- **Purchase Details:** Item Purchased, Category, Purchase Amount (USD), Season, Size, Color
- **Shopping behaviour:** Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type

Data Quality:

- **Missing Values:** 37 missing entries in the *Review Rating* column

4. Exploratory Data Analysis using Python:

We began with data preparation and cleaning in Python:

- Data Loading: Imported the dataset using pandas.
- Initial Exploration: Used describe () for summary statistics.

```
know all the columns summery statistic [by default u will get numeric statics but if u want to know catagory column as well use include=all]
.describe(include='all')
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900	3
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223	
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN	
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN	
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN	
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN	
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN	
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN	
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN	

Previous Purchases	Payment Method	Frequency of Purchases
900.000000	3900	3900
NaN	6	7
NaN	PayPal	Every 3 Months
NaN	677	584
25.351538	NaN	NaN
14.447125	NaN	NaN
1.000000	NaN	NaN
13.000000	NaN	NaN
25.000000	NaN	NaN
38.000000	NaN	NaN
50.000000	NaN	NaN

- **Missing Data Handling:** Identified null values and imputed missing entries in the review_rating column using the median rating for each product category.

- **Column Standardization:** Renamed all columns to snake_case to improve readability and maintain consistent documentation.

- **Feature Engineering:**
    - Created age_group column by binning customer ages.
    - Generated purchase_frequency_days column based on purchase data.

- **Data Consistency Check:** Verified redundancy between discount_applied and promo_code_used; removed promo_code_used to avoid duplication.

- **Database Integration:** Connected the Python script to MySQL and loaded the cleaned DataFrame into the database for subsequent SQL analysis.

5. Data Analysis using SQL (Business Transactions):

We performed structured analysis in **MySQL** to answer key business questions:

1. **Revenue by Gender:** Compared total revenue generated by male vs. female customers.

gender	revenue
Male	157890
Female	75191

2. **Revenue by Category:** Calculated total revenue by category.

Category	total_revenue
Clothing	104264
Accessories	74200
Footwear	36093
Outerwear	18524

3. **Top 5 Products by Rating:** Found products with the highest average review ratings.

item_purchased	avg_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.78

4. **Shipping Type Comparison:** Compared average purchase amounts between Standard and Express shipping.

shipping_type	avg_purchase_amount
Express	60.48
Standard	58.46

5. Subscribers vs. Non-Subscribers: Compared average spend and total revenue across subscription status.

subscription_status	total_customer	avg_spend	total_revenue
Yes	1053	59.49	62645
No	2847	59.87	170436

6. **Discount-Dependent Products:** Identified 5 products with the highest percentage of discounted purchases.

item_purchased	discount_percentage
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7. **Repeat Buyers & Subscriptions:** Checked whether customers with >5 purchases are more likely to subscribe.

subscription_status	repeted_buyers
Yes	958
No	2518

8. **Revenue by Age Group:** Calculated total revenue contribution of each age group.

age_group	total_revenue
Young Adults	62143
Middle Aged	59197
Adults	55978
Senior	55763

6. Dashboard in Power BI:

The dashboard includes KPIs for average purchase amount, average review rating, and customer count, along with visual analyses by category, age group, and subscription status.

7. Business Recommendations:

- Boost Subscriptions: Promote exclusive benefits for subscribers.
- Customer Loyalty Programs: Reward repeat buyers to move them into the “Loyal” segment.
- Review Discount Policy: Balance sales boosts with margin control.
- Product Positioning: Highlight top-rated and best-selling products in campaigns.
- Targeted Marketing : Focus efforts on high-revenue age groups and express-shipping users.